

BLAKE FARMAN

CURRICULUM VITAE

MATHEMATICS AND STATISTICS
LOUISIANA TECH UNIVERSITY
BFARMAN@LATECH.EDU
[HTTPS://WWW.BLAKEFARMAN.PHD](https://www.blakefarman.phd)

CONTACT INFORMATION

Nethken Hall
P.O. Box 10348
Ruston, Louisiana 71272
+1 318 257 2951

CITIZENSHIP

United States

RESEARCH INTERESTS

Algebra, (Noncommutative) Algebraic Geometry, Derived Categories, and Category Theory

EDUCATION

<i>Doctor of Philosophy</i> , Mathematics University of South Carolina, Columbia, SC Dissertation: Geometry of Derived Categories on Noncommutative Projective Schemes Advisor: Matthew Ballard	May 12, 2018
<i>Master of Science</i> , Mathematics University of Vermont, Burlington, VT Major Concentration: Abstract Algebra Minor Concentration: Algebraic Number Theory	May 22, 2011
<i>Bachelor of Science</i> , Computer Science Rensselaer Polytechnic Institute, Troy, NY Minors: Applied Mathematics and French	May 16, 2009

APPOINTMENTS

<i>Assistant Professor</i> Louisiana Tech University, Ruston, LA	September 2024 – Present
<i>Assistant Professor</i> University of Louisiana at Monroe, Monroe, LA	August 2020 – August 2024
<i>Capital One Endowed Professorship in Mathematics</i> University of Louisiana at Monroe, Monroe, LA	August 2020 – May 2023
<i>Visiting Assistant Professor</i> Lafayette College, Easton, PA	July 2018 – July 2020
<i>Graduate Teaching Assistant</i> University of South Carolina, Columbia, SC	August 2012 – May 2018
<i>Graduate Teaching Assistant</i> University of Vermont, Burlington, VT	August 2009 – May 2011

LEADERSHIP AND GOVERNANCE

<i>Communication Coordinator (Officer)</i> Louisiana-Mississippi Section Mathematical Association of America	February 2025 – Present
--	-------------------------

<i>Membership Outreach Director (Officer)</i> Board of Directors Friends of Black Bayou Black Bayou National Wildlife Refuge Monroe, LA	January 2025 – Present
---	------------------------

<i>Director at Large</i> Board of Directors Friends of Black Bayou Black Bayou National Wildlife Refuge Monroe, LA	January 2024 – January 2025
--	-----------------------------

EMPLOYMENT

<i>Associate</i> Princeton Consultants, Inc, Princeton, NJ Information Technology consultant.	August 2011 – August 2012
---	---------------------------

PUBLICATIONS

Ballard, M., Farman, B., 2021. Kernels for noncommutative projective schemes. *J. Noncommut. Geom.* 15, 1129–1180. <https://doi.org/10.4171/jncg/427>

Clifton, A., Farman, B., 2026. WIP: Using Career Panels to Spark Curiosity and Career Awareness Among Mathematics Majors. To Appear.

Farman, B., Clifton, A., Long, W., Cruse, K., 2025. Fostering Growth Mindsets: Implementing Standards-Based Grading in College Algebra, in: 2025 ASEE Annual Conference & Exposition Proceedings. Presented at the 2025 ASEE Annual Conference & Exposition, ASEE Conferences, Montreal, Quebec, Canada, p. 56592. <https://doi.org/10.18260/1-2--56592>

Clifton, A., Fendley, M., Farman, B., Coriell, J., 2025. Work in Progress: First-Year Engineering Students' Confidence in Communicating Mathematical Content, in: 2025 ASEE Annual Conference & Exposition Proceedings. Presented at the 2025 ASEE Annual Conference & Exposition, ASEE Conferences, Montreal, Quebec, Canada, p. 57495. <https://doi.org/10.18260/1-2--57495>

FORMALIZATION

Peer-reviewed contributions to the `Mathlib` library:

Introduced torsion theories for abelian categories. <code>CategoryTheory/Abelian/TorsionTheory</code> ; PR #36744.	Work In Progress
---	------------------

Closure properties of orthogonal complements and S. E. Dickson's theorem. <code>CategoryTheory/ObjectProperty</code> ; PR #44226.	Work In Progress
--	------------------

Short exact sequences from pulling back a subobject. <code>CategoryTheory/Abelian</code> ; PR #44229.	Work In Progress
--	------------------

Kernels of pullbacks of subobjects. <code>CategoryTheory/Subobject</code> ; PR #44223.	Work In Progress
---	------------------

Introduced and characterized radicals on abelian categories. <code>CategoryTheory/Abelian/Preradical</code> ; Commit 09d467d	March 16, 2026
---	----------------

Added Stenström's colon construction for preradicals. <code>CategoryTheory/Abelian/Preradical</code> ; Commit c505a69	March 6, 2026
--	---------------

Introduced the basic theory of preradicals on abelian categories. <code>CategoryTheory/Abelian/Preradical</code> ; Commit 1f308dd	February 26, 2026
--	-------------------

Defined the topologies associated to ideal filters. <code>RingTheory/IdealFilter</code> ; Commit a91e88c	February 2, 2026
Introduced ideal filters and Gabriel filters on the lattice of ideals of a ring. <code>RingTheory/IdealFilter</code> ; Commit 78484be	January 22, 2026
Generalized <code>Submodule.colon</code> to arbitrary subsets and upgraded the surrounding API. <code>RingTheory/Ideal</code> ; Commit 95093e9	January 16, 2026

FUNDING

<i>Formalizing the dg-Enhancement of Triangulated Categories</i> Anthropic AI for Science Six-month award of Claude usage credits to formalize dg-categories and the derived category of a dg-category in Lean's <code>Mathlib</code> library. Role: Principal Investigator (PI).	\$20,000
<i>Applied eXploration In Open Mathematics Lab [AXIOM Lab]</i> Louisiana Board of Regents Support Fund (LEQSF(2025-26)-ENH-DE-07) Targeted Departmental Enhancement Grant to establish a dedicated research collaboration lab within the Mathematics and Statistics program at Louisiana Tech University. Role: Principal Investigator (PI).	\$60,875
<i>Student Activity Enhancement Fee</i> Student Development, University of Louisiana at Monroe SAEF Grant to take mathematics students to the annual meeting of the Louisiana-Mississippi section of the Mathematical Association of America in New Orleans, LA.	\$2,000
<i>Capital One Endowed Professorship in Mathematics (2022 - 2023)</i>	\$5,370
<i>Capital One Endowed Professorship in Mathematics (2021 - 2022)</i>	\$5,070
<i>Capital One Endowed Professorship in Mathematics (2020 - 2021)</i>	\$4,790

SUBMITTED GRANTS

<i>Formalizing the Recognition Theorem for Cofibrantly Generated Model Categories in Lean: Foundations for Homotopy Theory and Derived Algebraic Geometry (Declined)</i> Research Competitiveness Subprogram Louisiana Board of Regents	\$69,110
<i>LEAPS-MPS: Building the Foundations of Formal (Declined)</i> <i>Homotopy Theory through Model Categories (FY 25)</i> National Science Foundation	\$165,753
<i>Fourier-Mukai Partners in Noncommutative Algebraic Geometry (Declined)</i> DMS - Algebra and Number Theory National Science Foundation	\$118,624
<i>Fourier-Mukai Partners in Noncommutative Algebraic Geometry (Declined)</i> Research Competitiveness Subprogram, FY 23 Louisiana Board of Regents Support Fund	\$113,918
<i>Fourier-Mukai Partners in Noncommutative Algebraic Geometry (Declined)</i> Research Competitiveness Subprogram, FY 24 Louisiana Board of Regents Support Fund	\$142,041

Derived Categories in Noncommutative Projective Algebraic Geometry \$210,000
(Declined)

Proposed NSF Mathematical Sciences Post Doctoral Research Fellowship to work with Ragnar Buchweitz (University of Toronto). Later revised to work with Michael Weems (University of Glasgow)

Dean's Dissertation Fellowship Nominee (Declined) \$25,000

Nominated by the Department of Mathematics at the University of South Carolina for a fellowship from the College of Arts and Sciences to spend a year in residence at the Institute of Advanced Study in Princeton, NJ.

AWARDS AND HONORS

Education Award 2025

College of Engineering and Science
Louisiana Tech University

Project NExT (New Experiences In Teaching) Fellow Red '22

The Mathematical Association of America

Rensselaer Medalist (2004) \$60,000

Scholarship to Rensselaer Polytechnic Institute.

GRADUATE MENTORING AND RESEARCH SUPERVISION

Joshua Bush 2025–2026

Ph.D. Student (Committee Chair), Computational Analysis and Modeling
Louisiana Tech University

Josiah Norman Graduating Spring 2026

M.S. Student (Committee Member)
Louisiana Tech University

UNDERGRADUATE MENTORING AND RESEARCH SUPERVISION

Joshua Robinson Spring 2025

Louisiana Tech University

Co-mentored with Ann Clifton on a Cayley graph research project, *Structural and Topological Properties of Cayley Graphs*, presented at the Louisiana Tech Undergraduate Research & Service-Learning Symposium and awarded second place in the Student Paper Competition at the LA/MS Section meeting of the Mathematical Association of America.

Riya Shrestha Spring 2024

Emerging Scholars Program, University of Louisiana at Monroe
Knot Theory research project focused on identifying a family of knots for which there is a winning strategy in the Region Unknot Game.

The Entrepreneurship Pelican Cup Spring 2023

Immerse AR

Served as faculty advisor for one of six finalists in the Pelican Cup Entrepreneurship business competition.

Sagar Bhetuwal Spring 2021

Emerging Scholars Program, University of Louisiana at Monroe

Number Theory research project focused on providing a formal verification of the proof of the Prime Number Theorem using the Lean proof assistant.

TEACHING INNOVATIONS

Mathematical Art Installation

Spring 2025

With Ann Clifton and Glen Whitney (Studio Infinity), guided students in building a compound of five tetrahedra from road-cone barriers and radio-antenna mounting hardware, exhibited alongside the senior design showcase. Recognized, together with the AXIOM Lab, by the 2025 COES Education Award.

TEACHING EXPERIENCE

LOUISIANA TECH UNIVERSITY

Course Number	Course Name	Term
Math 5843	Algebraic Geometry	S26
Math 587	Interactive Theorem Proving with Lean	S25
Math 5083	Intro to Abstract Algebra	F25
Math 4083	Intro to Abstract Algebra	F25
Math 490	Interactive Theorem Proving with Lean	S25
Math 4903	Algebraic Geometry	S26
Math 308	Introduction to Linear Algebra	W25
Math 243	Calculus III	F24

UNIVERSITY OF LOUISIANA AT MONROE

Course Number	Course Name	Section
Math 4007	History of Mathematics	44692 (F22)
Math 3086	Modern Algebra	44367 (F21), 63395 (S23)
Math 3080	Real Analysis	64863 (S22), 44174 (F23)
Math 3073	Numerical Methods	45407 (F23)
Math 3040	Introduction to Algebra and Analysis	64086 (S24)
Math 2040	Foundations of Mathematics	65041 (S23)
Math 2032	Calculus III	44879 (F22), 44879 (F24)
Math 2002	Applied Linear Algebra	41001 (S21), 64042 (F21), 64042 (S23)
Math 1032	Calculus II	40999 (F24)
Math 1031	Calculus I	63789 (S24), 40997 (F24)
Math 1018	Contemporary Mathematics	40992, 40994 (F21), 60847, 62225 (S22)
Math 1011	College Algebra	42181, 43002, 43013 (F20), 62690, 62691 (S21), 63057, 63167 (S24)
Math 1007	Contemporary Mathematics	45282, 45288 (F22), 64979 (S23), 45282, 45288 (F23)

LAFAYETTE COLLEGE

Course Number	Course Name	Sections
Math 272	Linear Algebra with Applications	01 (S20)
Math 162	Calculus II	06, 07 (S19), 01, 03 (F19)
Math 161	Calculus I	01, 04 (F18), 04 (F19), 01, 02 (S20)
Math 161	Summer Program to Advance Leadership	SPAL (Su19)

UNIVERSITY OF SOUTH CAROLINA

Course Number	Course Name	Sections
Math 170	Finite Mathematics	007 (S14), 007 (S16)
Math 142	Calculus II	004 (Su14), 020 (S18)
Math 122	Calculus for Business Administration and Social Sciences	E01 (S17)
Math 116	Brief Precalculus Mathematics	E1B, E2B (F13), E1B, E2B (F15), E1B, E2B (F16)
Math 115	Precalculus Mathematics	011 (F14)
Math 111	Basic College Mathematics	M13 (Su13), 006 (S15), 006 (F17)

UNIVERSITY OF VERMONT

Course Number	Course Name	Sections
Math 019	Fundamentals of Calculus I	S, W(F10)
Math 017	Applications of Finite Math	G (F09), A (S10), Z2 (Su10)

PRESENTATIONS

<i>Formal Methods in Modern Algebra</i>	October 9, 2025
Computational Analysis and Modeling Seminar Louisiana Tech University, Ruston, LA	
<i>A Little Algebraic Topology</i>	April 11, 2025
Faculty Seminar COES Graduate Student Council Louisiana Tech University, Ruston, LA	
<i>Fostering Growth Mindsets: Implementing Standards-Based Grading in College Algebra</i>	February 22, 2025
Annual Meeting of the Louisiana/Mississippi Section Mathematical Association of America Jackson, MS	
<i>Algebra, Geometry, and Categories, Oh My!</i>	May 11, 2023
Louisiana Tech University Ruston, LA	
<i>Implementing Standards Based Grading in STEM</i>	March 28, 2023
University of Louisiana System For our Future Conference 2023 Lake Charles, LA	
<i>Algebra, Geometry, and Categories, Oh My!</i>	March 11, 2020
Department Seminar, Lafayette College, Easton, PA	
<i>A Little Abstract Nonsense</i>	May 21, 2018
Lafayette College, Easton, PA	
<i>Abstract Nonsense: Grokking Category Theory</i>	May 17, 2018
Wilkes University, Wilkes-Barre, PA	
<i>Kernels for Noncommutative Projective Schemes</i>	November 28, 2017
Graduate Colloquium University of South Carolina, Columbia, SC	
<i>Kernels for Noncommutative Projective Schemes</i>	October 12, 2017
Algebra Seminar University of New Brunswick, Fredericton, NB, CA	
<i>Kernels for Noncommutative Projective Schemes</i>	October 9, 2017
Algebra Seminar Temple University, Philadelphia, PA	

<i>Kernels for Noncommutative Projective Schemes</i> Algebraic Geometry Seminar University of South Carolina, Columbia, SC	August 28, 2017
<i>Noncommutative Projective Schemes</i> Temple Graduate Student Conference in Algebra, Geometry, and Topology Temple University, Philadelphia, PA	May 13, 2016
<i>Noncommutative Projective Schemes</i> Student Algebra/Number Theory Seminar University of South Carolina, Columbia, SC	May 14, 2016
<i>Fully Faithful Abstract Nonsense:</i> <i>Representable Functors and The Yoneda Embedding</i> Graduate Colloquium University of South Carolina, Columbia, SC	November 2015
<i>Fully Faithful Abstract Nonsense:</i> <i>Representable Functors and The Yoneda Embedding</i> Graduate Student Algebraic Geometry, Commutative Algebra, and Number Theory Seminar University of South Carolina, Columbia, SC	July 2015

CONFERENCES AND WORKSHOPS

<i>LA/MS Sectional Meeting</i> Mathematical Association of America Jackson, MS	February 20 – February 22, 2025
<i>Math Corequisite Academy Workshop</i> Louisiana Board of Regents Baton Rouge, LA	October 15, 2024
<i>LA/MS Sectional Meeting</i> Mathematical Association of America New Orleans, LA	February 29 – March 2, 2024
<i>High School Math Teacher and Faculty Collaboration</i> Louisiana Department of Education Baton Rouge, LA	November 30, 2023
<i>Math Corequisite Academy Workshop</i> Louisiana Board of Regents Baton Rouge, LA	November 7, 2023
<i>Mathematical Association of America MathFest</i> Tampa, FL	August 1–6, 2023
<i>Mathematics Research Community</i> Derived Categories, Arithmetic and Geometry, Bridgeland Stability Group Beaver Hollow Conference Center, Java, NY	June 4–10, 2023
<i>Joint Mathematics Meetings</i> Boston, MA	January 3–7, 2023
<i>Mathematical Association of America MathFest</i> Philadelphia, PA	August 3–6, 2022
<i>Mathematical Association of America Project NExT (New Experiences in Teaching)</i> Philadelphia, PA	August 1–3, 2022
<i>Series on Exploring Combinatorics and Number Theory (SECANT) IV</i> Cedar Crest College, Allentown, PA (Online)	January 15–16, 2022

<i>Derived, Birational, and Categorical Algebraic Geometry</i> Invited Participant Banff International Research Station, Banff, Alberta, CA	November 14–19, 2021
<i>Series on Exploring Combinatorics and Number Theory (SECANT) III</i> Cedar Crest College, Allentown, PA (Online)	January 15–16, 2021
<i>Derived, Birational, and Categorical Algebraic Geometry</i> Invited Participant Banff International Research Station, Banff, Alberta, CA (Online)	November 2–6, 2020
<i>Joint Mathematics Meetings</i> Denver, CO	January 15–17, 2020
<i>Series on Exploring Combinatorics and Number Theory (SECANT) II</i> Cedar Crest College, Allentown, PA	October 12–13, 2019
<i>Joint Mathematics Meetings</i> Baltimore, MD	January 16–19, 2019
<i>Series on Exploring Combinatorics and Number Theory (SECANT)</i> Cedar Crest College, Allentown, PA	November 17–18, 2018
<i>Georgia Algebraic Geometry Symposium</i> Georgia Institute of Technology, Atlanta, GA	February 23–25, 2018
<i>Workshop on Topics in Algebraic Geometry</i> The University of North Carolina at Chapel Hill, Chapel Hill, NC	November 3–5, 2017
<i>Palmetto Number Theory Series (PANTS) XXVIII</i> University of Tennessee, Knoxville, TN	September 16–17, 2017
<i>Georgia Algebraic Geometry Symposium</i> University of Georgia, Athens, GA	March 3–5, 2017
<i>Workshop on Symplectic Varieties and Geometric Representation Theory</i> The University of North Carolina at Chapel Hill, Chapel Hill, NC	October 28–30, 2016
<i>UGA-SWAG: Summer Workshop in Algebraic Geometry</i> University of Georgia, Athens, GA	August 26–29, 2016
<i>Temple Graduate Student Conference in Algebra, Geometry, and Topology</i> Temple University, Philadelphia, PA	May 14–15, 2016
<i>Georgia Algebraic Geometry Symposium</i> Emory University, Atlanta, GA	October 2015
<i>AMS Summer Institute on Algebraic Geometry</i> University of Utah, Salt Lake City, UT	July 27–31, 2015
<i>Georgia Algebraic Geometry Symposium</i> University of Georgia, Athens, GA	October 2014
<i>Arizona Winter School</i> University of Arizona, Tucson, AZ	March 2014
<i>Palmetto Number Theory Series (PANTS) XXI</i> Clemson University, Clemson, SC	December 2013
<i>Palmetto Number Theory Series (PANTS) XX</i> Davidson College, Davidson, NC	September 2013
<i>Algebraic Geometry Northeastern Series (AGNES)</i> Yale University, New Haven, CT	April 2013
<i>AMS Spring Western Sectional Meeting</i> University of Colorado, Boulder, CO	April 2013
<i>Palmetto Number Theory Series (PANTS) XIX</i> University of South Carolina, Columbia, SC	December 2012
<i>Palmetto Number Theory Series (PANTS) XVIII</i> Wake Forest University, Winston-Salem, NC	September 2012

PROFESSIONAL MEMBERSHIP

<i>Mathematical Association of America</i>	Member
<i>American Mathematical Society</i>	Member

SERVICE

TO THE PROFESSION

<i>Section NExT Committee Member</i> Louisiana-Mississippi Section Mathematical Association of America	February 2025 – Present
--	-------------------------

TO THE STATE

<i>Mindset Meauxtivor</i> Louisiana Board of Regents	Fall 2024 – Spring 2025
---	-------------------------

TO THE UNIVERSITY

LOUISIANA TECH UNIVERSITY

<i>Faculty Senate</i> College of Engineering and Science Representative; member of the subcommittee revising Policy 1301	2025–2026
<i>Undergraduate Advising</i> Advisees: J. D. Vessel and Bryce Labbie	2025–2026
<i>Career Panels for Mathematics Majors</i> Collaborated with Ann Clifton and the Mathematics Club to organize career panels for mathematics majors.	2025–2026
<i>Literary Rally</i> Proctor	March 15, 2025
<i>Undergraduate Research & Service-Learning Symposium</i> Judge	March 14, 2025
<i>Professors on the Plaza</i>	October 16, 2024

UNIVERSITY OF LOUISIANA AT MONROE

<i>Browse on the Bayou</i> Mathematics program representative for ULM's Fall on-campus recruiting event	September 30, 2022
<i>Mathematics Hiring Committee Chair</i>	Fall 2023
<i>Mathematics Hiring Committee Member</i>	Fall 2022 – Spring 2023
<i>Co-Requisite Course Design</i> Contemporary Mathematics	Spring 2022
<i>Browse on the Bayou</i> Mathematics program representative for ULM's Spring on-campus recruiting event	March 21, 2022
<i>Investiture Ceremony Marshal</i>	October 1, 2021
<i>University Curriculum Committee Member</i>	2021–2024
<i>Louisiana Math Forward Committee Member</i>	2021
<i>Mathematics Hiring Committee Member</i>	Spring 2021

UNIVERSITY OF SOUTH CAROLINA
High School Math Contest Proctor

2013–2015

TO THE COMMUNITY

National Hunting and Fishing Day

September 27, 2025

Represented Friends of Black Bayou at the Louisiana Department of Wildlife and Fisheries event for National Hunting and Fishing day.
 Black Bayou Lake National Wildlife Refuge

Fall Celebration

October 19, 2024

Setup, Operated Canoe/Kayak Rental, Breakdown
 Black Bayou Lake National Wildlife Refuge

Water Sweep

September 14, 2024

Waterway cleanup at Black Bayou Lake National Wildlife Refuge
 Ouachita Green

ORGANIZER

Student Algebra/Number Theory Seminar

Summer 2016

University of South Carolina

Student Algebraic Geometry, Commutative Algebra, and Number Theory Seminar

Summer 2015

University of South Carolina

Student Algebraic Geometry Seminar

Spring 2015

University of South Carolina

Student Algebraic Geometry Seminar

Fall 2014

University of South Carolina